Statistics 6910, Section 003, Spring 2013, Quiz 3 (25 Points)

Thursday, January 31, 2013

Your Name: ____________________

Question 1: Arrays and Apply (**6 Points**)

The array `arr` is created as follows in R:

```
arr = array(1:7, c(2, 4, 3))
```

This means, `arr` has _______ columns, _______ panels, and _______ rows. (1 point)
For each of the following 4 R expressions, indicate the resulting output. Read carefully!

```
#1 (2 points)
arr
```

```
#2 (1 point)
arr[-1, -(1:2), -(1:2)]
```

```
#3 (1 point)
arr[c(TRUE, FALSE), , c(FALSE, TRUE, FALSE)]
```

```
#4 (1 point)
apply(arr, 3, sum)
```

Question 2: Functions (**10 Points**)

For many statistical calculations, we need the following sums of 2 vectors $(x_1, \ldots, x_n)$ and $(y_1, \ldots, y_n)$:

$$\sum_{i=1}^{n} x_i, \quad \sum_{i=1}^{n} x_i^2, \quad \sum_{i=1}^{n} y_i, \quad \sum_{i=1}^{n} y_i^2, \quad \sum_{i=1}^{n} x_i y_i.$$

Write a function called `xySums` that calculates these 5 sums and returns a named list (consisting of `SumX, SumX2, SumY, SumY2, SumXY`) as its return value.

`xySums` takes two vectors `x` and `y` as its arguments. Neither of these has a default value.

There is **no need** for error checking, i.e., if the vectors are of different lengths or if they are not numeric (or integer), simply let R report a default error message.

Note that

```
> (1:10) * (1:10)
 [1]   1   4   9  16  25  36  49  64  81 100
```

My function is given below:

Question 3: Earthquakes and Apply (**9 Points**)

When dealing with eqarthquakes, we are usually interested in their locations, and not only in their magnitudes. In this question, we will focus on earthquakes that occurred in the San Francisco area.

Below are a few expressions and results that should help to remind you on the structure of the earthquake data:

```
CAquakes = read.table(file = "http://www.consrv.ca.gov/cgs/rghm/quakes/Documents/ms49epicenters.txt",
                      header = TRUE)

> dim(CAquakes)
[1] 383   4
>
> class(CAquakes)
[1] "data.frame"
>
> class(CAquakes$Latitude)
[1] "numeric"
>
> class(CAquakes$Longitude)
[1] "numeric"
>
> head(CAquakes)
      Date Latitude Longitude   M
1 18001011     36.8    -121.5 5.5
2 18001122     32.9    -117.8 6.3
3 18030000     34.2    -118.1 5.5
4 18030525     32.8    -117.1 5.5
5 18060325     34.4    -119.7 5.5
6 18080621     37.8    -122.6 5.5
>
> tail(CAquakes)
        Date Latitude Longitude   M
378 19951228  38.7210 -119.6590 5.5
379 19970122  40.2720 -124.3940 5.6
380 19981127  40.6670 -125.3840 5.6
381 19991016  34.6815 -116.2848 5.8
382 19991016  34.6000 -116.2700 7.1
383 19991016  34.4400 -116.2500 5.7
>
> inSFArea = (CAquakes$Latitude > 36) & (CAquakes$Latitude < 39) &
             (CAquakes$Longitude > -124) & (CAquakes$Longitude < -120)
>
> head(inSFArea)
[1]  TRUE FALSE FALSE FALSE FALSE  TRUE
```

The map (see Figure 1) has been created in R with the following commands (for now, just ignore these commands — but feel free to experiment with them at home):

```
library(RgoogleMaps)

bb = qbbox(c(min(CAquakes$Latitude), max(CAquakes$Latitude)),c(min(CAquakes$Longitude), max(CAquakes$Longitude)))

MyMap = GetMap.bbox(bb$lonR, bb$latR, destfile = "CAquakes.png", GRAYSCALE = TRUE)

tmp = PlotOnStaticMap(MyMap, lat = CAquakes$Latitude, lon = CAquakes$Longitude,
  cex = 1.5, pch = 20, col = "red", add = FALSE, axes = TRUE, mar = c(3, 3, 0, 0))

tmp = PlotOnStaticMap(MyMap, lat = c(36, 36, 39, 39, 36), lon = c(-124, -120, -120, -124, -124),
  col = "black", add = TRUE, FUN = lines, lwd = 2)
```

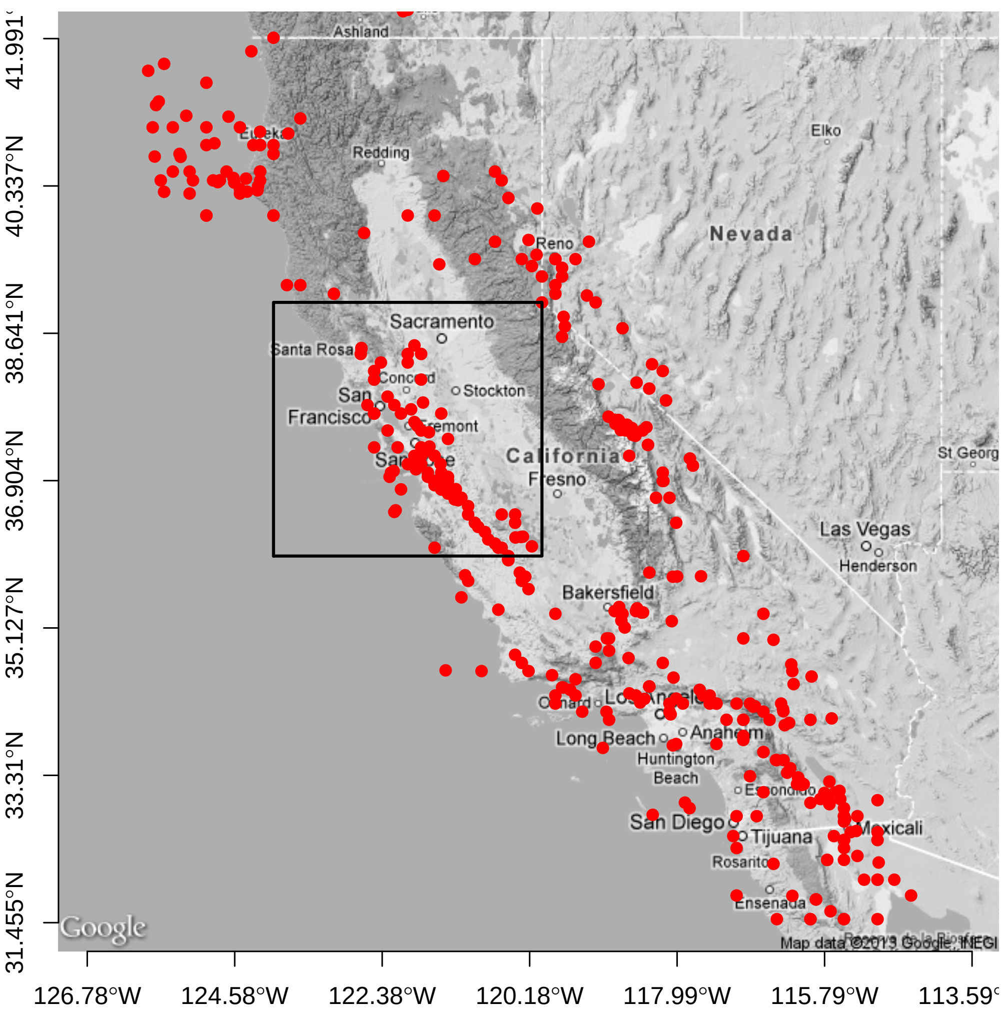

Figure 1: Location of earthquakes in California from 1800 to 1999. The area of interest around San Francisco is inside the rectangular area.

For this question, assume that we are primarily interested in earthquakes that occurred in the San Francisco area. This is the rectangular area (strictly) inside the black box in Figure 1. The lower left corner for this area is (Latitude, Longitude) $= (36, -124)$ and the upper right corner is (Latitude, Longitude) $= (39, -120)$. Recall that negative values for longitude represent locations west of the Greenwich Meridian, i.e., a longitude of -120 alternatively can be written as 120°W.

A logical vector (of length 383), called `inSFArea` already has been created! This vector contains the information whether the epicenter of an earthquake was (strictly) inside the area specified above, i.e., it is `TRUE` if (Latitude, Longitude) is (strictly) inside this area and `FALSE` otherwise.

Use a function from the `apply` family in **all** of your answers below (this can be a different one for each answer). The `inSFArea` vector may also be useful for some of your answers. As usual, be as efficient as possible when you indicate your **single** R expression for each part.

1. What was the median magnitude in the area of interest and what was the median magnitude outside the area of interest? The result should be a vector of length 2.

2. Someone is interested in the geographic center of all the earthquakes recorded in this data file. You may disagree that this is a good measure to summarize the spatial locations, but nevertheless, this person insists that you calculate this center point. So, what was the average (= mean) latitude and the average (= mean) longitude of **all** the earthquakes? The result should be a vector of length 2.

3. How many of the recorded 383 earthquakes were recorded in the 19th century (1800–1899) and how many were recorded in the 20th century (1900–1999)? The result should be a vector of length 2.